Sam Schlesinger

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SamuelSchlesinger.github.io
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Education

2019 **B.A. Computer Science and Math**, *UMass*, Amherst

Skills

Cryptography, Security, Distributed Systems, Rust, Haskell, TypeScript/JavaScript, Internet Standards, Web Servers, PostgreSQL, Kubernetes, Smart Contracts

Professional Experience

2022–Present **Senior Software Engineer**, *Google Chrome AI Security*

- Tech lead developing private authentication protocols for the web using zero-knowledge proofs.
- Implemented a private zero-knowledge banking protocol.
- Co-authored the Anonymous Tokens with Hidden Metadata standards drafts and Rust implementation; deployed to protect billions of users from spam texts via RCS/iMessage integration.
- Co-authored the Anonymous Credit Tokens (ACT) standards drafts and Rust implementation for unlinkable, partially spendable API credits.
- Led a three-day cross-browser PACT workshop producing three Moderation of unLinkable Endorsements (MoLE) Internet-Drafts; built an end-to-end Chromium prototype; presented at IETF 126. BoF planned for IETF 127 in San Francisco.
- Designed the Private Proof API for anti-fraud in a post-third-party-cookie web. Led three engineers to implement in Chrome. Drafted novel anonymous credential primitives.
- Co-chair of W3C Anti-Fraud Community Group.

2022 **Software Engineer / Scientist**, *Casper Labs*

- Co-designed novel consensus algorithm (presented at FOCODILE 2022). Implemented in Casper node, deployed in 2.0 upgrade.
- Researched zero-knowledge use cases and proposed network optimizations.
- Consulted C-suite on network security and scalability strategy.

2019–2022 **Software Engineer (Team Lead)**, *SimSpace*

- Designed and implemented Attack Designer program.
- Designed consensus-lite protocol for multi-datacenter cluster coordination.
- Backend security lead: identified and fixed cryptographic vulnerabilities.

Community & Outreach

Creator Computable Secrets, a YouTube channel explaining cryptography, formal methods, and theoretical computer science.

Reviewer CSLib, reviewing formalized computer science contributions to its Lean library.